STAT 2020 Discussion
Week 4

Shuo Li

The University of Iowa
Department of Statistics and Actuarial Science

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Announcements

- Please make sure you know your **section number**
- Please **bring a calculator** to the quiz
- If you have questions, feel free to email me

This Week's Main Skill

- This week focuses on two key ideas:
  - **Law of Total Probability**
  - **Bayes' Rule**
- Main goal: learn how to **set up** probability problems
- Most mistakes come from **incorrect setup**, not calculation

Today we will practice the problem-solving recipe.

Notation Reminder

- X: random variable
- x: a possible value of X
- Example:

$$P(X = x)$$

- CDF definition:

$$F(x) = P(X \leq x)$$

We will revisit these ideas later.

Formal Definition of Limit (ε-δ)

We say

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Rightarrow \quad |f(x) - L| < \varepsilon.$$

- ε controls how close $f(x)$ is to L
- δ controls how close x is to a

Understanding CDF Limits Using the Definition

Recall the definition:

$$\lim_{x\to\infty} F(x) = 1$$

means: for every $\varepsilon > 0$, there exists M such that

$$x > M \quad \Rightarrow \quad |F(x) - 1| < \varepsilon.$$

Interpretation:

- No matter how small ε is (e.g. $\varepsilon = 0.01$),
- we can find a large enough cutoff M,
- so that when x is larger than M, $F(x)$ is within ε of 1.

Similarly,

$$\lim_{x\to-\infty} F(x) = 0$$

means: for every $\varepsilon > 0$, there exists M such that

$$x < M \quad \Rightarrow \quad |F(x) - 0| < \varepsilon.$$

Partition (The Key Setup)

- Many probability problems become easier if we split into cases.
- Events $E_1, E_2, \ldots, E_k$ form a **partition** if:

$$E_i \cap E_j = \varnothing \quad (i \neq j), \qquad \bigcup_{i=1}^{k} E_i = S$$

- In words:
  - cases do not overlap
  - cases cover the whole sample space

Law of Total Probability

If $E_1, \ldots, E_k$ form a partition, then

$$P(A) = \sum_{i=1}^{k} P(E_i)P(A \mid E_i)$$

- Think of it as a **weighted average** of conditional probabilities
- Common mistake: forgetting the weights $P(E_i)$

Bayes' Rule

If $E_1, \ldots, E_k$ form a partition, then

$$P(E_i \mid A) = \frac{P(E_i)P(A \mid E_i)}{\sum_{j=1}^{k} P(E_j)P(A \mid E_j)}$$

- Bayes' Rule is used for **backward reasoning**
- We observe A and want to infer which case E_i is likely

Common Mistakes to Avoid

- Mixing up conditional probabilities:

$$P(A \mid B) \neq P(B \mid A)$$

- Forgetting the denominator in Bayes' Rule
- Choosing cases that do NOT form a partition
  - if your cases do not add up to 1, setup is wrong
- Confusing **mutually exclusive** with **independent**

Mutually Exclusive vs Independent

Mutually Exclusive:

$$A \cap B = \varnothing$$

They cannot happen at the same time.

Independent:

$$P(A \cap B) = P(A)P(B)$$

Knowing one happened does not change the probability of the other.

These are completely different concepts.

Example 1: Total Probability Practice

A campus shuttle has two possible routes:

- Route 1: taken with probability 0.6
- Route 2: taken with probability 0.4

Let A be the event that the shuttle arrives late.

$$P(A \mid R_1) = 0.10, \qquad P(A \mid R_2) = 0.30$$

Question: Compute $P(A)$.

Solution: Example 1

Step 1: Identify the partition.

R_1, R_2 form a partition.

Solution: Example 1

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$$R_1, R_2 \text{ form a partition.}$$

Step 2: Apply total probability.

$$P(A) = P(R_1)P(A \mid R_1) + P(R_2)P(A \mid R_2)$$

Solution: Example 1

Step 1: Identify the partition.

$$R_1, R_2 \text{ form a partition.}$$

Step 2: Apply total probability.

$$P(A) = P(R_1)P(A \mid R_1) + P(R_2)P(A \mid R_2)$$

$$P(A) = 0.6(0.10) + 0.4(0.30) = 0.06 + 0.12 = 0.18$$

$$\boxed{P(A) = 0.18}$$

Example 2: Bayes' Rule Practice

A spam filter classifies emails as spam.

- $P(S) = 0.20$ (email is actually spam)
- $P(S^c) = 0.80$ (email is not spam)

The filter performance:

$$P(+ \mid S) = 0.90 \qquad P(+ \mid S^c) = 0.10$$

Let $+$ mean "flagged as spam."
Question: Find $P(S \mid +)$.

Solution: Example 2 (Bayes)

We want:

$$P(S \mid +) = \frac{P(S)P(+ \mid S)}{P(+)}$$

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We want:

$$P(S \mid +) = \frac{P(S)P(+ \mid S)}{P(+)}$$

First compute the denominator using total probability:

$$P(+) = P(S)P(+ \mid S) + P(S^c)P(+ \mid S^c)$$

Solution: Example 2 (Bayes)

We want:

$$P(S \mid +) = \frac{P(S)P(+ \mid S)}{P(+)}$$

First compute the denominator using total probability:

$$P(+) = P(S)P(+ \mid S) + P(S^c)P(+ \mid S^c)$$

$$P(+) = 0.2(0.9) + 0.8(0.1) = 0.18 + 0.08 = 0.26$$

Solution: Example 2 (Bayes)

We want:

$$P(S \mid +) = \frac{P(S)P(+ \mid S)}{P(+)}$$

First compute the denominator using total probability:

$$P(+) = P(S)P(+ \mid S) + P(S^c)P(+ \mid S^c)$$

$$P(+) = 0.2(0.9) + 0.8(0.1) = 0.18 + 0.08 = 0.26$$

Now apply Bayes:

$$P(S \mid +) = \frac{0.2(0.9)}{0.26} = \frac{0.18}{0.26} \approx 0.692$$

$$\boxed{P(S \mid +) \approx 0.692}$$

Example 3: Medical Test (Bayes + Total Probability)

A disease affects 1% of a population.

A test result is **positive** $(+)$ with:

$$P(+ \mid D) = 0.95 \qquad P(+ \mid D^c) = 0.08$$

where D means the person has the disease.

Question: If a person tests positive, what is $P(D \mid +)$?

Solution: Example 3

We apply Bayes' Rule:

$$P(D \mid +) = \frac{P(D)P(+ \mid D)}{P(+)}$$

Solution: Example 3

We apply Bayes' Rule:

$$P(D \mid +) = \frac{P(D)P(+ \mid D)}{P(+)}$$

First compute $P(+)$ using total probability:

$$P(+) = P(D)P(+ \mid D) + P(D^c)P(+ \mid D^c)$$

Solution: Example 3

We apply Bayes' Rule:

$$P(D \mid +) = \frac{P(D)P(+ \mid D)}{P(+)}$$

First compute $P(+)$ using total probability:

$$P(+) = P(D)P(+ \mid D) + P(D^c)P(+ \mid D^c)$$

Given:

$$P(D) = 0.01, \quad P(D^c) = 0.99$$

Solution: Example 3

We apply Bayes' Rule:

$$P(D \mid +) = \frac{P(D)P(+ \mid D)}{P(+)}$$

First compute $P(+)$ using total probability:

$$P(+) = P(D)P(+ \mid D) + P(D^c)P(+ \mid D^c)$$

Given:

$$P(D) = 0.01, \quad P(D^c) = 0.99$$

So:

$$P(+) = 0.01(0.95) + 0.99(0.08) = 0.0095 + 0.0792 = 0.0887$$

Solution: Example 3

We apply Bayes' Rule:

$$P(D \mid +) = \frac{P(D)P(+ \mid D)}{P(+)}$$

First compute $P(+)$ using total probability:

$$P(+) = P(D)P(+ \mid D) + P(D^c)P(+ \mid D^c)$$

Given:

$$P(D) = 0.01, \quad P(D^c) = 0.99$$

So:

$$P(+) = 0.01(0.95) + 0.99(0.08) = 0.0095 + 0.0792 = 0.0887$$

Now compute the posterior:

$$P(D \mid +) = \frac{0.01(0.95)}{0.0887} = \frac{0.0095}{0.0887} \approx 0.107$$

Wrap-Up: Key Takeaways

- Always identify the **partition (cases)** first
- Total Probability:

$$P(A) = \sum_i P(E_i)P(A \mid E_i)$$

- Bayes' Rule:

$$P(E_i \mid A) = \frac{P(E_i)P(A \mid E_i)}{P(A)}$$

- Common mistakes:
  - confusing $P(A \mid B)$ and $P(B \mid A)$
  - missing the denominator

In-Class Attendance Check (Quick): Compute $P(A)$

Suppose events E_1 and E_2 form a partition of the sample space.

Given:

$$P(E_1) = 1, \quad P(E_2) = 0,$$

$$P(A \mid E_1) = 0.10, \quad P(A \mid E_2) = 0.10.$$

- Compute $P(A)$.

Please submit: Full Name; Section Number; Your Answer

In-Class Attendance Check (Quick): Compute $P(A)$

Suppose events E_1 and E_2 form a partition of the sample space.

Given:

$$P(E_1) = 1, \quad P(E_2) = 0,$$

$$P(A \mid E_1) = 0.10, \quad P(A \mid E_2) = 0.10.$$

- Compute $P(A)$.

Please submit: Full Name; Section Number; Your Answer

Answer:

$$P(A) = P(E_1)P(A \mid E_1) + P(E_2)P(A \mid E_2) = 0.10.$$